In [ ]:

```
EXPERIMENT - 4
```

In [ ]:

```
DATA MANIPULATION WITH PANDAS
```

In [ ]:

```
Step 1 — Install and Import Pandas Library
```

In [1]:

```
# Importing Pandas
import pandas as pd
```

In [ ]:

```
Step 2 — Importing the Dataset
```

In [ ]:

```
# Importing the Dataset
df_241UDS0092 = pd.read_csv(
    "Experiment_4_241UDS0092_Retail_Sales_Data.csv"
)
# Reading the head of the Dataset
df_241UDS0092.head(5)
```

Out[2]: Columns 1–8 of 11

	Product_ID	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent
0	P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0
1	P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0
2	P1003	Water Bottle	Home	Visakhapatnam	31	480	8.0
3	P1004	USB-C Cable	Electronics	Pune	58	390	12.0
4	P1005	Desk Lamp	Home	Mumbai	26	1250	7.0

Columns 9–11 of 11 · first column repeated

	Customer_Rating	Stock_Available	Payment_Method
0	4.2	18	UPI
1	4.5	35	Cash
2	4.1	22	Card
3	4.4	41	UPI
4	4.0	14	Card

In [ ]:

```
Step 3 — Set Index for the DataFrame
```

In [ ]:

```
# Set Index for the DataFrame
df_241UDS0092 = df_241UDS0092.set_index("Product_ID")
df_241UDS0092.head()
```

Out[3]: Columns 1–8 of 10

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0	4.5
P1003	Water Bottle	Home	Visakhapatnam	31	480	8.0	4.1
P1004	USB-C Cable	Electronics	Pune	58	390	12.0	4.4
P1005	Desk Lamp	Home	Mumbai	26	1250	7.0	4.0

Columns 9–10 of 10 · first column repeated

	Stock_Available	Payment_Method
Product_ID		
P1001	18	UPI
P1002	35	Cash
P1003	22	Card
P1004	41	UPI
P1005	14	Card

In [ ]:

```
Step 4 — Info of the DataFrame
```

In [ ]:

```
# Info of the DataFrame
df_241UDS0092.info()
```

```
<class 'pandas.DataFrame'>
Index: 30 entries, P1001 to P1030
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Product_Name      30 non-null     str
 1   Category          30 non-null     str
 2   City              30 non-null     str
 3   Units_Sold        30 non-null     int64
 4   Unit_Price        30 non-null     int64
 5   Discount_Percent  30 non-null     float64
 6   Customer_Rating   30 non-null     float64
 7   Stock_Available   30 non-null     int64
 8   Payment_Method    30 non-null     str
dtypes: float64(2), int64(3), str(4)
memory usage: 2.3+ KB
```

In [ ]:

```
Step 5 — Columns & Rows Selection
```

In [ ]:

```
# Selecting a Single Column
df_241UDS0092["Product_Name"]
```

Out[5]:

```
Product_ID
P1001       Wireless Mouse
P1002         Notebook Set
P1003         Water Bottle
P1004          USB-C Cable
P1005            Desk Lamp
P1006             Pen Pack
P1007    Bluetooth Speaker
P1008           Coffee Mug
P1009             Backpack
P1010             Keyboard
P1011          File Folder
P1012      Table Organizer
P1013          Phone Stand
P1014     Highlighter Pack
P1015           Power Bank
P1016            Lunch Box
P1017         Sticky Notes
P1018        Laptop Sleeve
P1019            LED Strip
P1020        Cushion Cover
P1021           Marker Set
P1022            Earphones
P1023           Wall Clock
P1024         Travel Pouch
P1025           Calculator
P1026           Smart Plug
P1027          Storage Box
P1028      Document Folder
P1029          Phone Pouch
P1030             Mini Fan
Name: Product_Name, dtype: str
```

In [ ]:

```
Step 6 — Selecting Multiple Columns
```

In [ ]:

```
# Selecting Multiple Columns
        df_241UDS0092[
    ["Product_Name", "Category", "City"]
].head(3)
```

Out[7]:

	Product_Name	Category	City
Product_ID			
P1001	Wireless Mouse	Electronics	Hyderabad
P1002	Notebook Set	Stationery	Vijayawada
P1003	Water Bottle	Home	Visakhapatnam

In [ ]:

```
Step 7 — Select Row Using iloc
```

In [ ]:

```
# Select by row index using iloc (position-based)
df_241UDS0092.iloc[1]
```

Out[8]:

```
Product_Name        Notebook Set
Category              Stationery
City                  Vijayawada
Units_Sold                    75
Unit_Price                   220
Discount_Percent            10.0
Customer_Rating              4.5
Stock_Available               35
Payment_Method              Cash
Name: P1002, dtype: object
```

In [ ]:

```
Step 8 — Select Row Using loc
```

In [ ]:

```
# Select by row index using loc (label-based)
df_241UDS0092.loc["P1030"]
```

Out[9]:

```
Product_Name           Mini Fan
Category            Electronics
City                  Hyderabad
Units_Sold                   19
Unit_Price                 1750
Discount_Percent           15.0
Customer_Rating             4.5
Stock_Available              10
Payment_Method              UPI
Name: P1030, dtype: object
```

In [ ]:

```
Step 9 — Manipulating Data Based on Columns
```

In [ ]:

```
# Display first two rows
df_241UDS0092.head(2)
```

Out[10]:

Columns 1–8 of 10

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0	4.5

Columns 9–10 of 10 · first column repeated

	Stock_Available	Payment_Method
Product_ID		
P1001	18	UPI
P1002	35	Cash

In [ ]:

```
Step 10 — Creating Total Sales Column
```

In [ ]:

```
# Creating Total Sales Column

df_241UDS0092["Total_Sales_241UDS0092"] = (
    df_241UDS0092["Units_Sold"] *
    df_241UDS0092["Unit_Price"]
)

df_241UDS0092.head(2)
```

Out[12]: Columns 1–8 of 11

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0	4.5

Columns 9–11 of 11 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110
Product_ID			
P1001	18	UPI	27300
P1002	35	Cash	16500

In [ ]:

```
Step 11 — Creating Discount Amount Column
```

In [ ]:

```
# Creating Discount Amount Column
df_241UDS0092["Discount_Amount_241UDS0092"] = (
    df_241UDS0092["Total_Sales_241UDS0092"] *
    df_241UDS0092["Discount_Percent"] / 100
)
df_241UDS0092.head(2)
```

Out[13]: Columns 1–8 of 12

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0	4.5

Columns 9–12 of 12 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110
Product_ID				
P1001	18	UPI	27300	1365.0
P1002	35	Cash	16500	1650.0

In [ ]:

```
Step 12 — Creating Net Sales Column
```

In [ ]:

```
# Creating Net Sales Column
df_241UDS0092["Net_Sales_241UDS0092"] = (
    df_241UDS0092["Total_Sales_241UDS0092"] -
    df_241UDS0092["Discount_Amount_241UDS0092"]
)
df_241UDS0092.head(2)
```

Out[14]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0	4.5

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1001	18	UPI	27300	1365.0	25935.0
P1002	35	Cash	16500	1650.0	14850.0

In [ ]:

```
Step 13 — Round Calculated Values
```

In [ ]:

```
# Round calculated columns to 2 decimal places
df_241UDS0092["Discount_Amount_241UDS0092"] = (
    df_241UDS0092["Discount_Amount_241UDS0092"].round(2)
)
df_241UDS0092["Net_Sales_241UDS0092"] = (
    df_241UDS0092["Net_Sales_241UDS0092"].round(2)
)
df_241UDS0092.head(2)
```

Out[15]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0	4.5

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1001	18	UPI	27300	1365.0	25935.0
P1002	35	Cash	16500	1650.0	14850.0

In [ ]:

```
Step 14 — Conditional Filtering
```

In [ ]:

```
# Products having Units Sold greater than 50
df_241UDS0092[
    df_241UDS0092["Units_Sold"] > 50
].head(2)
```

Out[16]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0	4.5
P1004	USB-C Cable	Electronics	Pune	58	390	12.0	4.4

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1002	35	Cash	16500	1650.0	14850.0
P1004	41	UPI	22620	2714.4	19905.6

In [ ]:

```
Step 15 — Products with High Sales AND Low Stock
```

In [ ]:

```
# Products with Total Sales > 30000 AND Stock Available < 25
df_241UDS0092[
    (df_241UDS0092["Total_Sales_241UDS0092"] > 30000) &
    (df_241UDS0092["Stock_Available"] < 25)
].head(3)
```

Out[17]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1005	Desk Lamp	Home	Mumbai	26	1250	7.0	4.0
P1007	Bluetooth Speaker	Electronics	Bangalore	33	1850	6.0	4.3
P1009	Backpack	Accessories	Hyderabad	29	1450	11.0	4.7

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1005	14	Card	32500	2275.0	30225.0
P1007	19	UPI	61050	3663.0	57387.0
P1009	16	UPI	42050	4625.5	37424.5

In [ ]:

```
Step 16 — Products from Hyderabad OR Mumbai
```

In [ ]:

```
# Products from Hyderabad OR Mumbai
df_241UDS0092[
    (df_241UDS0092["City"] == "Hyderabad") |
    (df_241UDS0092["City"] == "Mumbai")
].head()
```

Out[18]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1005	Desk Lamp	Home	Mumbai	26	1250	7.0	4.0
P1009	Backpack	Accessories	Hyderabad	29	1450	11.0	4.7
P1013	Phone Stand	Accessories	Mumbai	51	620	13.0	4.0
P1016	Lunch Box	Home	Hyderabad	46	720	7.0	4.2

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1001	18	UPI	27300	1365.0	25935.0
P1005	14	Card	32500	2275.0	30225.0
P1009	16	UPI	42050	4625.5	37424.5
P1013	31	Card	31620	4110.6	27509.4
P1016	28	Cash	33120	2318.4	30801.6

In [ ]:

```
Step 17 — Products with Rating ≥ 4.5 AND Stock < 30
```

In [ ]:

```
# Products with high customer rating and low stock
df_241UDS0092[
    (df_241UDS0092["Customer_Rating"] ≥ 4.5) &
    (df_241UDS0092["Stock_Available"] < 30)
]
```

Out[19]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1009	Backpack	Accessories	Hyderabad	29	1450	11.0	4.7
P1012	Table Organizer	Home	Pune	39	540	10.0	4.5
P1015	Power Bank	Electronics	Chennai	24	2100	9.0	4.6
P1018	Laptop Sleeve	Accessories	Delhi	35	1150	14.0	4.8
P1024	Travel Pouch	Accessories	Delhi	41	760	11.0	4.6
P1026	Smart Plug	Electronics	Mumbai	27	1550	13.0	4.7
P1030	Mini Fan	Electronics	Hyderabad	19	1750	15.0	4.5

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1009	16	UPI	42050	4625.5	37424.5
P1012	21	UPI	21060	2106.0	18954.0
P1015	12	Card	50400	4536.0	45864.0
P1018	17	Card	40250	5635.0	34615.0
P1024	24	Cash	31160	3427.6	27732.4
P1026	13	Card	41850	5440.5	36409.5
P1030	10	UPI	33250	4987.5	28262.5

In [ ]:

```
Step 18 — Filtering Multiple Cities Using .isin()
```

In [ ]:

```
# Filtering with Multiple Cities using .isin()
df_241UDS0092[
    df_241UDS0092["City"].isin(
        ["Hyderabad", "Bangalore", "Pune"]
    )
].head()
```

Out[20]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1004	USB-C Cable	Electronics	Pune	58	390	12.0	4.4
P1007	Bluetooth Speaker	Electronics	Bangalore	33	1850	6.0	4.3
P1009	Backpack	Accessories	Hyderabad	29	1450	11.0	4.7
P1012	Table Organizer	Home	Pune	39	540	10.0	4.5

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1001	18	UPI	27300	1365.0	25935.0
P1004	41	UPI	22620	2714.4	19905.6
P1007	19	UPI	61050	3663.0	57387.0
P1009	16	UPI	42050	4625.5	37424.5
P1012	21	UPI	21060	2106.0	18954.0

In [ ]:

```
Step 19 — Filtering Multiple Categories Using .isin()
```

In [ ]:

```
# Filtering with Multiple Categories using .isin()
df_241UDS0092[
    df_241UDS0092["Category"].isin(
        ["Electronics", "Home"]
    )
].head()
```

Out[21]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1003	Water Bottle	Home	Visakhapatnam	31	480	8.0	4.1
P1004	USB-C Cable	Electronics	Pune	58	390	12.0	4.4
P1005	Desk Lamp	Home	Mumbai	26	1250	7.0	4.0
P1007	Bluetooth Speaker	Electronics	Bangalore	33	1850	6.0	4.3

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1001	18	UPI	27300	1365.0	25935.0
P1003	22	Card	14880	1190.4	13689.6
P1004	41	UPI	22620	2714.4	19905.6
P1005	14	Card	32500	2275.0	30225.0
P1007	19	UPI	61050	3663.0	57387.0

In [ ]:

```
Step 20 — Filtering Multiple Payment Methods Using .isin()
```

In [ ]:

```
# Filtering with Multiple Payment Methods using .isin()
df_241UDS0092[
    df_241UDS0092["Payment_Method"].isin(
        ["UPI", "Card"]
    )
].head()
```

Out[22]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyderabad	42	650	5.0	4.2
P1003	Water Bottle	Home	Visakhapatnam	31	480	8.0	4.1
P1004	USB-C Cable	Electronics	Pune	58	390	12.0	4.4
P1005	Desk Lamp	Home	Mumbai	26	1250	7.0	4.0
P1007	Bluetooth Speaker	Electronics	Bangalore	33	1850	6.0	4.3

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1001	18	UPI	27300	1365.0	25935.0
P1003	22	Card	14880	1190.4	13689.6
P1004	41	UPI	22620	2714.4	19905.6
P1005	14	Card	32500	2275.0	30225.0
P1007	19	UPI	61050	3663.0	57387.0

In [ ]:

```
Step 21 — Sorting Table
```

In [ ]:

```
# Sorting Table based on single column - Units Sold (Descending)
df_241UDS0092.sort_values(
    "Units_Sold",
    ascending=False
).head()
```

Out[23]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1006	Pen Pack	Stationery	Delhi	96	150	15.0	4.6
P1017	Sticky Notes	Stationery	Vijayawada	91	130	5.0	4.1
P1011	File Folder	Stationery	Delhi	82	180	8.0	4.3
P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0	4.5
P1028	Document Folder	Stationery	Chennai	74	210	8.0	4.4

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1006	60	Cash	14400	2160.0	12240.0
P1017	55	UPI	11830	591.5	11238.5
P1011	48	Cash	14760	1180.8	13579.2
P1002	35	Cash	16500	1650.0	14850.0
P1028	46	UPI	15540	1243.2	14296.8

In [ ]:

```
Step 22 — Sort by Unit Price — Ascending
```

In [ ]:

```
# Sorting based on Unit Price - Ascending
df_241UDS0092.sort_values(
    "Unit_Price",
    ascending=True
).head()
```

Out[24]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1017	Sticky Notes	Stationery	Vijayawada	91	130	5.0	4.1
P1006	Pen Pack	Stationery	Delhi	96	150	15.0	4.6
P1011	File Folder	Stationery	Delhi	82	180	8.0	4.3
P1014	Highlighter Pack	Stationery	Bangalore	67	190	6.0	4.4
P1028	Document Folder	Stationery	Chennai	74	210	8.0	4.4

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1017	55	UPI	11830	591.5	11238.5
P1006	60	Cash	14400	2160.0	12240.0
P1011	48	Cash	14760	1180.8	13579.2
P1014	39	UPI	12730	763.8	11966.2
P1028	46	UPI	15540	1243.2	14296.8

In [ ]:

```
Step 23 — Sorting Using Multiple Columns
```

In [ ]:

```
# Sorting based on Category and Units Sold
df_241UDS0092.sort_values(
    ["Category", "Units_Sold"],
    ascending=[True, False]
).head()
```

Out[25]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1029	Phone Pouch	Accessories	Vijayawada	56	330	10.0	4.0
P1013	Phone Stand	Accessories	Mumbai	51	620	13.0	4.0
P1024	Travel Pouch	Accessories	Delhi	41	760	11.0	4.6
P1018	Laptop Sleeve	Accessories	Delhi	35	1150	14.0	4.8
P1009	Backpack	Accessories	Hyderabad	29	1450	11.0	4.7

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1029	37	Card	18480	1848.0	16632.0
P1013	31	Card	31620	4110.6	27509.4
P1024	24	Cash	31160	3427.6	27732.4
P1018	17	Card	40250	5635.0	34615.0
P1009	16	UPI	42050	4625.5	37424.5

In [ ]:

```
Step 24 — Updating Data in the Table
```

In [ ]:

```
# Updating Hyderabad as Hyd in City Column
df_241UDS0092["City"] = df_241UDS0092["City"].replace(
    ["Hyderabad"],
    ["Hyd"]
)
df_241UDS0092[
    df_241UDS0092["City"] == "Hyd"
].head(2)
```

Out[26]: Columns 1–8 of 13

	Product_Name	Category	City	Units_Sold	Unit_Price	Discount_Percent	Customer_Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyd	42	650	5.0	4.2
P1009	Backpack	Accessories	Hyd	29	1450	11.0	4.7

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Net_Sales_241UDS0110
Product_ID					
P1001	18	UPI	27300	1365.0	25935.0
P1009	16	UPI	42050	4625.5	37424.5

In [ ]:

```
Step 25 — Updating a Specific Product Value
```

In [ ]:

```
# Updating Product P1001 Unit Price
df_241UDS0092.loc[
    "P1001",
    "Unit_Price"
] = 700
print(
    df_241UDS0092.loc[
        "P1001",
        "Unit_Price"    ]
)
```

```
700
```

In [ ]:

```
Step 26 — Changing Column Names
```

In [ ]:

```
# Changing Column Names
df_241UDS0092 = df_241UDS0092.rename(
    columns={
        "Product_Name": "Item_Name",
        "City": "Location",
        "Customer_Rating": "Rating",
        "Net_Sales_241UDS0092": "Revenue"
    }
)
df_241UDS0092.head(1)
```

Out[28]: Columns 1–8 of 13

	Item_Name	Category	Location	Units_Sold	Unit_Price	Discount_Percent	Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyd	42	700	5.0	4.2

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Revenue
Product_ID					
P1001	18	UPI	27300	1365.0	25935.0

In [ ]:

```
Step 27 — Final DataFrame
```

In [ ]:

```
# Display the final DataFrame
df_241UDS0092.head()
```

Out[29]: Columns 1–8 of 13

	Item_Name	Category	Location	Units_Sold	Unit_Price	Discount_Percent	Rating
Product_ID							
P1001	Wireless Mouse	Electronics	Hyd	42	700	5.0	4.2
P1002	Notebook Set	Stationery	Vijayawada	75	220	10.0	4.5
P1003	Water Bottle	Home	Visakhapatnam	31	480	8.0	4.1
P1004	USB-C Cable	Electronics	Pune	58	390	12.0	4.4
P1005	Desk Lamp	Home	Mumbai	26	1250	7.0	4.0

Columns 9–13 of 13 · first column repeated

	Stock_Available	Payment_Method	Total_Sales_241UDS0110	Discount_Amount_241UDS0110	Revenue
Product_ID					
P1001	18	UPI	27300	1365.0	25935.0
P1002	35	Cash	16500	1650.0	14850.0
P1003	22	Card	14880	1190.4	13689.6
P1004	41	UPI	22620	2714.4	19905.6
P1005	14	Card	32500	2275.0	30225.0

In [ ]:

```
Step 28 — Final Dataset Information
```

In [ ]:

```
# Final information of the DataFrame
df_241UDS0092.info()
```

```
<class 'pandas.DataFrame'>
Index: 30 entries, P1001 to P1030
Data columns (total 12 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Item_Name                   30 non-null     str
 1   Category                    30 non-null     str
 2   Location                    30 non-null     str
 3   Units_Sold                  30 non-null     int64
 4   Unit_Price                  30 non-null     int64
 5   Discount_Percent            30 non-null     float64
 6   Rating                      30 non-null     float64
 7   Stock_Available             30 non-null     int64
 8   Payment_Method              30 non-null     str
 9   Total_Sales_241UDS0110      30 non-null     int64
 10  Discount_Amount_241UDS0110  30 non-null     float64
 11  Revenue                     30 non-null     float64
dtypes: float64(4), int64(4), str(4)
memory usage: 4.1+ KB
```

In [ ]:

```
RESULT:
By using the Pandas library, various data manipulation techniques such as selecting specific rows and columns, rounding and updating values, creating calculated
columns, conditional filtering, sorting, and renaming columns were successfully implemented on the retail sales dataset.
```